

Multilingual Speech Classification and Clustering System

Machine Learning Course Final Project

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1 Introduction

This project aims to develop a machine learning system capable of identifying the spoken language from audio recordings. The dataset comprises voice samples from four languages: German, Italian, Korean, and Spanish. Our approach involves a comprehensive pipeline including data cleaning, feature extraction, supervised classification, and unsupervised clustering.

In the supervised learning phase, we evaluate the performance of Support Vector Machines (SVM), Random Forest, and Multi-Layer Perceptrons (MLP) to determine the most effective model for this task. We use metrics such as Accuracy, Precision, Recall, and F1-Score, along with Confusion Matrices, to rigorously evaluate these models.

In the unsupervised learning phase, we employ K-Means and Hierarchical Clustering algorithms to discover natural groupings within the data without using language labels. We analyze the quality of these clusters using Silhouette Scores and Purity metrics. Additionally, we utilize t-SNE (t-Distributed Stochastic Neighbor Embedding) for dimensionality reduction to visualize the high-dimensional feature space in 2D.

To put it simply, we are teaching a computer to listen to a short audio clip and guess whether the person is speaking German, Italian, Korean, or Spanish. We do this by breaking down the sound into mathematical numbers (features) and training the computer to recognize patterns in these numbers. We also try to see if the computer can group these languages on its own without being told which is which (clustering).

2 Data Cleaning and Feature Extraction

Audio data in its raw form often contains noise and silence that can hinder model performance. Just like a human might struggle to understand speech in a noisy room or if there's a long pause before someone speaks, a computer model faces similar challenges. To address this, we implemented the following preprocessing steps:

2.1 Silence Trimming

Raw recordings often contain leading or trailing silence, or pauses between sentences. These silent segments do not carry linguistic information and introduce unnecessary dimensionality and noise. By removing them using the `librosa.effects.trim` function (from the `librosa` library), we ensure the model focuses purely on the active speech segments.

2.2 Data Augmentation

The original dataset consisted of 720 samples, distributed evenly across the four languages (180 samples per language). To prevent overfitting and improve model robustness, we implemented data augmentation by injecting random Gaussian noise (factor=0.005) into the original audio signals. This effectively doubled our dataset size to 1440 samples, with 360 samples per language (180 original + 180 augmented).

Training complex models on small datasets often leads to overfitting, where the model memorizes specific samples rather than learning generalizable features. Augmentation forces the model to learn robust features that are invariant to minor background noise.

2.3 Feature Extraction and Storage

Raw audio waveforms are high-dimensional arrays and unsuitable for direct input into standard classifiers. We processed the audio at a sample rate of 22050 Hz. We extracted 39 specific features that characterize the timbre, pitch, and spectral content of speech. These features were aggregated into a structured dataset and saved as a CSV file for efficient processing.

2.3.1 Feature Selection

- **MFCCs (Mel-Frequency Cepstral Coefficients):** We extracted the Mean and Variance of 13 MFCCs (26 features).
 - **Technical:** MFCCs approximate the human auditory system's response. They are the standard for speech recognition as they capture the "timbre" or spectral envelope of the sound.
 - **Simple Terms:** Think of "Timbre" as the "texture" or "color" of a sound. It's what makes a guitar sound different from a piano even if they play the exact same note. For speech, MFCCs help the computer distinguish the unique sounds (phonemes like "aa", "ee", "sh") that make up different languages.
- **Spectral Centroid:** (1 feature, Mean).
 - **Technical:** This measures the "center of mass" of the spectrum, corresponding to the perceived "brightness" of a sound.
 - **Simple Terms:** This measures how "sharp" or "dull" a voice sounds. A high value means the sound is "bright" and "hissy" (like a lot of 's' sounds), while a low value means it is "deep" and "muffled" (like 'm' or 'u' sounds). Some languages naturally use more sharp sounds than others.
- **Chroma Feature:** (12 features, Mean).
 - **Technical:** Projects the spectrum onto 12 pitch classes.
 - **Simple Terms:** This relates to the musical notes or "pitch" of the voice. Even though people aren't singing, every language has a specific melody or rhythm (intonation). For example, Italian is often described as "sing-songy," while other languages might be more monotone. this feature helps the computer hear that "melody."

3 Classification

We utilized the `scikit-learn` library to implement and evaluate three distinct classifiers. The loaded dataset was split into training and testing sets using a `GroupShuffleSplit` strategy (80% Train / 20% Test). This resulted in approximately 1152 training samples and 288 testing samples. Crucially, grouping by 'Student ID' ensures that all clips from a single speaker are either in the train or test set, preventing the model from recognizing the speaker instead of the language.

We scaled the data using `StandardScaler` to normalize features (mean=0, variance=1). Algorithms like Support Vector Machines (SVM) and Neural Networks (MLP) are sensitive to the magnitude of input features. For instance, Spectral Centroid values might range in the thousands (e.g., 2000Hz), while MFCC values might range clearly between -10 and 10. Without

scaling, the model would treat the specific "loudness" of the frequency (large numbers) as more important than the timbre (small numbers) simply because the numbers are bigger. Scaling puts everyone on a level playing field so the model compares apples to apples.

Since the number of features (39) is relatively low compared to the number of samples, we deemed dimensionality reduction techniques like PCA or LDA unnecessary for the classification phase, preferring to allow the models to learn from the raw feature interactions.

3.1 Model Architectures and Parameters

1. Support Vector Machine (SVM):

- **Configuration:** Kernel = 'RBF', C=1.0
- **Accuracy:** 83%
- **Why:** SVMs are excellent at finding complex, curved boundaries between groups in high-dimensional space. The RBF kernel allows it to separate languages that overlap significantly, but it struggles slightly compared to the MLP because audio patterns are often extremely complex.

2. Random Forest:

- **Configuration:** n_estimators = 100 (100 Decision Trees).
- **Accuracy:** 74%
- **Why:** Random Forests rely on splitting data using perpendicular cuts (like a flowchart). Audio features often have blurry, diagonal relationships that are hard for trees to capture efficiently. It's robust, but less sensitive to subtle acoustic nuances than the other models.

3. Multi-Layer Perceptron (MLP):

- **Configuration:** Hidden Layers = 4 layers of 1000 neurons, Max Iterations = 100,000.
- **Accuracy:** 87%
- **Why:** As a deep Neural Network, the MLP mimics the layered processing of the human brain. The large capacity (4 layers of 1000 neurons) allows it to learn hierarchical features (combining simple sounds into complex language patterns) making it the most accurate model for this task.

Each trained model was serialized and saved as a `.pk1` file for reproducibility and deployment.

4 Clustering

In the unsupervised phase, we blindly fed the audio data to algorithms (K-Means and Hierarchical Clustering) and asked: "Can you grouping these into 4 piles?" We did *not* tell them which language was which. This step helps us explore the natural structure of the data.

4.1 Optimal Number of Clusters

We used two methods to determine the optimal number of clusters (k). Although we expected $k = 4$ (for the 4 languages), the data suggested otherwise, consistently pointing to $k = 2$ as the strongest natural grouping.

Why $k=2$? The acoustic features (MFCCs, Pitch, Centroid) are heavily influenced by the speaker's gender (Male vs Female). The average pitch of a male voice (85-180 Hz) is significantly lower than a female voice (165-255 Hz). This creates a massive mathematical divide in the data that overshadows the subtle differences between languages. The clustering algorithms saw Gender first, Language second. However, we proceeded with $k = 4$ to evaluate how well the algorithms could recover the 4 language groups.

4.1.1 Elbow Method

We ran K-Means for $k = 1$ to 10 and plotted the inertia (sum of squared errors). As shown below, the "elbow" of the curve appears around $k = 2$, suggesting the strongest natural split is binary.

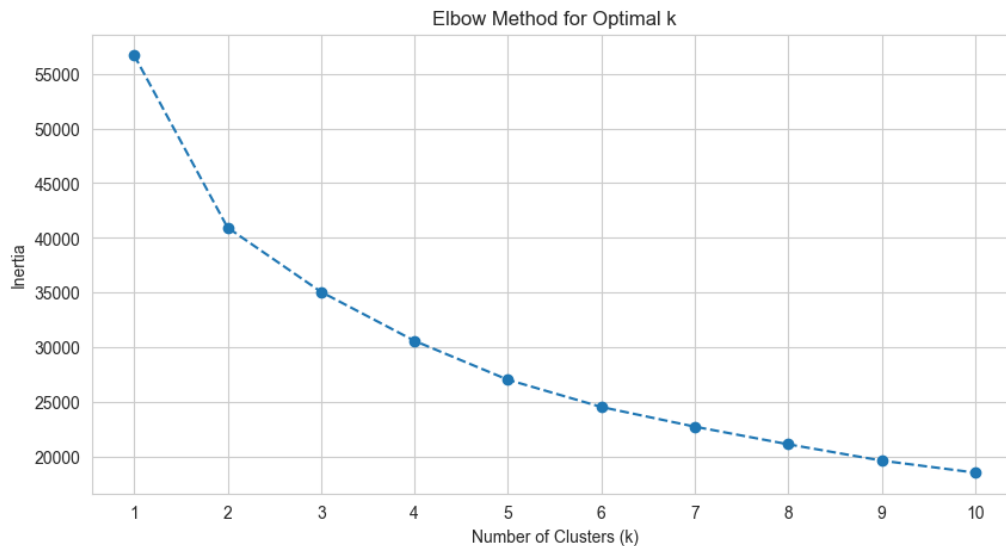


Figure 1: Elbow Method Plot showing optimal $k=2$

4.1.2 Silhouette Analysis

We calculated the Silhouette Score for $k = 2$ to 10. The score measures how well-separated the clusters are.

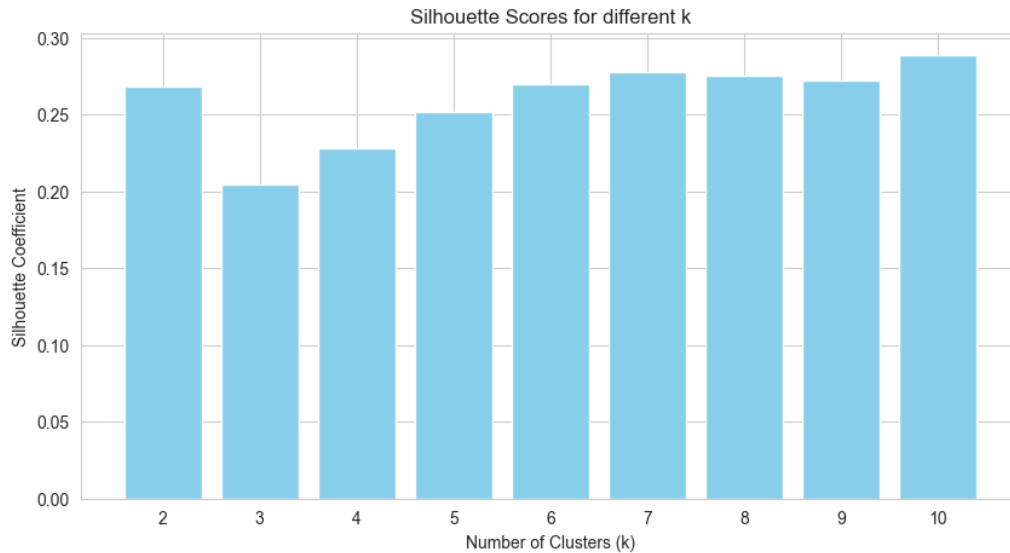


Figure 2: Silhouette Scores for different k values

The highest score was also found for $k = 2$.

4.2 Visualization (t-SNE)

We used t-SNE (t-Distributed Stochastic Neighbor Embedding) to project the 39-dimensional data into a 2D map for visualization. t-SNE is a non-linear dimensionality reduction technique that minimizes the divergence between two distributions: a distribution that measures pairwise similarities of the input objects and a distribution that measures pairwise similarities of the corresponding low-dimensional points in the embedding. It is particularly good at preserving local structure. Imagine you have a ball of yarn with 39 different colored threads tangled together. t-SNE is like carefully pulling those threads apart and laying them flat on a table so you can see which colors stick together and which ones are far apart, without breaking the connections between neighbors.

We visualized True Language Labels showing the actual ground truth, K-Means Clusters showing how K-Means grouped the data and, Hierarchical Clusters showing how Agglomerative clustering grouped the data.

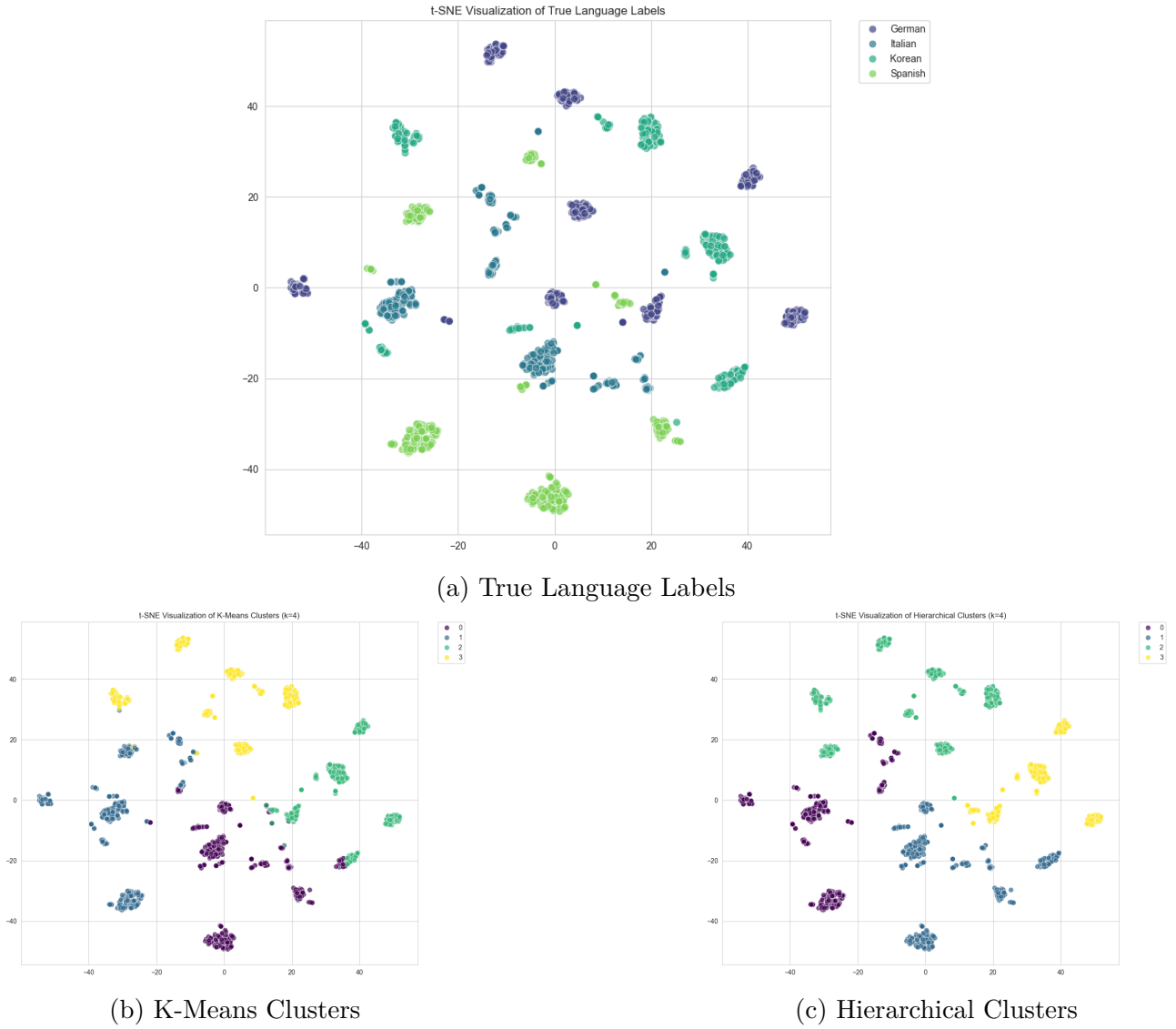


Figure 3: t-SNE Visualizations of Audio Features

The clustering algorithms (K-Means and Hierarchical) created clean, distinct partitions of data. They found a strong logical structure in the audio files. However, when we look at the True Labels plot, we see that the languages are mixed together within these partitions. This indicates that the strongest natural patterns are not the Languages themselves, but likely other acoustic properties like gender or recording environment. The algorithms found structure, just not the specific linguistic structure we hoped for unsupervised.

4.3 Clustering Algorithms

To investigate the natural groupings further, we computed the cluster composition tables for $k = 4$. These tables reveal how samples from each language were distributed across the clusters.

4.3.1 K-Means Cluster Composition

Language	Cluster 0	Cluster 1	Cluster 2	Cluster 3
German	42	43	138	137
Italian	172	169	8	11
Korean	56	31	124	149
Spanish	160	147	17	36

Table 1: Distribution of samples per Language in K-Means Clusters

4.3.2 Hierarchical Cluster Composition

Language	Cluster 0	Cluster 1	Cluster 2	Cluster 3
German	46	35	137	142
Italian	188	158	7	7
Korean	30	90	150	90
Spanish	100	150	83	27

Table 2: Distribution of samples per Language in Hierarchical Clusters

The tables confirm the hypothesis that the clusters are not aligning with languages. In K-Means, Cluster 0 and 1 captured almost all the Italian and Spanish samples, but also small chunks of German and Korean. Cluster 2 and 3 heavily favor German and Korean. The same pattern can be seen in Hierarchical clusters.

Why? This split (Romance languages vs Germanic/Asian languages) might actually be reflecting Gender or Recording Conditions more strongly than language family. If the German/Korean speakers were predominantly female higher pitch or recorded in similar acoustic environments, they would group together. The algorithm grouped similar-sounding files, but similar sounding usually means similar pitch (gender) rather than similar language phonemes.

5 Evaluation

5.1 Classification

We evaluated the models on the unseen test set (288 testing samples) to measure their ability to generalize to new speakers. By ensuring no speaker overlap between training and testing, the following results truly reflect the model’s ability to identify language patterns rather than memorizing voice characteristics.

5.1.1 Model Comparison

The table below summarizes the performance metrics for each model, separated into Weighted average (considering class imbalance) and Macro average (treating all classes equally):

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	0.74	0.84	0.74	0.75
SVM (RBF)	0.83	0.87	0.83	0.84
MLP	0.87	0.91	0.87	0.88

Table 3: Model Performance Comparison (Weighted Average)

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	0.74	0.77	0.83	0.77
SVM (RBF)	0.83	0.81	0.87	0.83
MLP	0.87	0.89	0.93	0.89

Table 4: Model Performance Comparison (Macro Average)

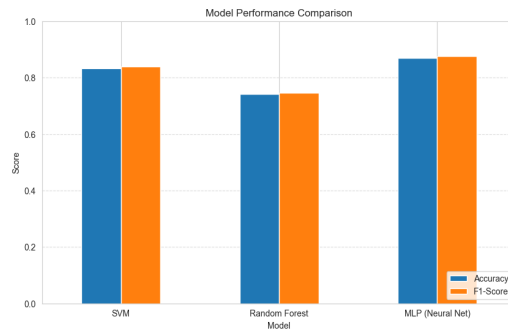


Figure 4: Performance Comparison of Classification Models

5.1.2 Model Analysis

We dive deeper into the performance of each model by examining their per-class metrics and confusion matrices.

Multi-Layer Perceptron (MLP) was the clear winner with an F1-Score of **0.88**. This suggests that the difference between languages isn't a simple straight line or a simple flowchart rule. It requires the complex, layered pattern recognition that only a neural network provides.

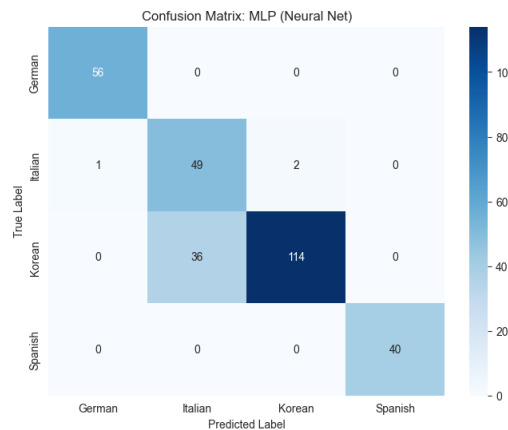


Figure 5: Confusion Matrix for MLP

Class	Precision	Recall	F1-Score	Support
German	0.98	1.00	0.99	56
Italian	0.58	0.94	0.72	52
Korean	0.98	0.76	0.86	150
Spanish	1.00	1.00	1.00	40
Accuracy			0.87	298
Macro Avg	0.89	0.93	0.89	298
Weighted Avg	0.91	0.87	0.88	298

Table 5: Detailed Classification Report: MLP (Neural Net)

Support Vector Machine (SVM) came in second (0.84). It shows that the data is separable, but perhaps the boundaries are fuzzier than the SVM would like.

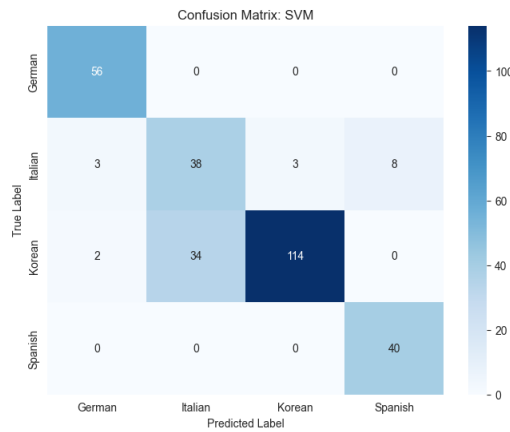


Figure 6: Confusion Matrix for SVM

Class	Precision	Recall	F1-Score	Support
German	0.92	1.00	0.96	56
Italian	0.53	0.73	0.61	52
Korean	0.97	0.76	0.85	150
Spanish	0.83	1.00	0.91	40
Accuracy			0.83	298
Macro Avg	0.81	0.87	0.83	298
Weighted Avg	0.87	0.83	0.84	298

Table 6: Detailed Classification Report: SVM

Random Forest struggled the most (0.75). Decision trees split data using perpendicular cuts. Audio data likely has diagonal or curved relationships between features that trees struggle to approximate efficiently without massive amounts of data.

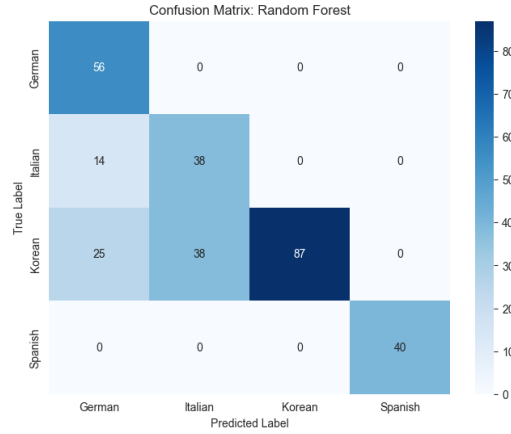


Figure 7: Confusion Matrix for Random Forest

Class	Precision	Recall	F1-Score	Support
German	0.59	1.00	0.74	56
Italian	0.50	0.73	0.59	52
Korean	1.00	0.58	0.73	150
Spanish	1.00	1.00	1.00	40
Accuracy			0.74	298
Macro Avg	0.77	0.83	0.77	298
Weighted Avg	0.84	0.74	0.75	298

Table 7: Detailed Classification Report: Random Forest

5.1.3 MLP Detailed Error Analysis

- **Perfect Success (German and Spanish):** The model achieved nearly perfect recall for German and Spanish. This implies these languages have very distinct acoustic signatures in our dataset. German has harsh consonants and unique vowel sounds that MFCCs pick up easily. Spanish has a consistent, rapid syllable timing that is distinct.
- **The "Italian Problem":** The primary weakness was identifying Italian. The precision was only 0.58. Why?
 1. **Linguistic Similarity:** Italian and Spanish are both Romance languages. They share similar phonemes and similar intonation patterns. The model likely confused some Italian clips for Spanish.
 2. **Dataset Imbalance or Variability:** Even though we balanced the classes, the quality or diversity of the Italian samples might be lower. If the Italian speakers had very different accents or recording environments compared to the others, the model might struggle to find a unifying "Italian" feature.
- **Korean:** Korean had good but not perfect performance. As the only non-European language, we expected it to be very distinct. Any confusion here might be due to the Chroma (pitch) features potentially overlapping with the intonation of other languages depending on the speaker's emotional tone.

5.2 Clustering

5.2.1 Silhouette Score and Purity

We evaluated the quality of our unsupervised models using two primary metrics.

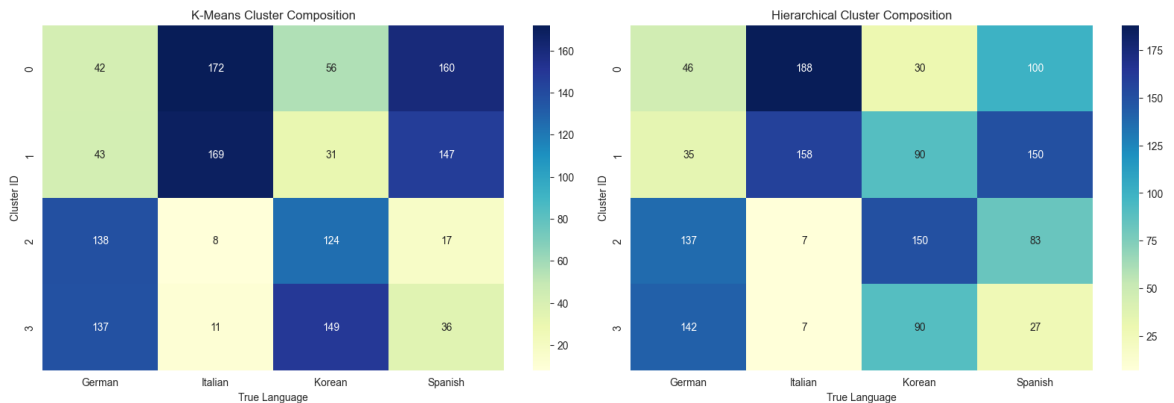
- **Silhouette Score:** Measures the mean distance between a sample and all other points in the same cluster (a) versus the mean distance to points in the nearest neighboring cluster (b). The score is given by $(b - a)/\max(a, b)$. Range: $[-1, 1]$. This asks: "Do I fit in well with my group, or would I be happier in the group next door?" A high score means the data point sits comfortably in the middle of its own cluster and is far away from other clusters.
- **Cluster Purity:** We assign each cluster to the class which is most frequent in the cluster, and then calculate the accuracy of this assignment. Imagine sorting a basket of mixed fruit blindly into 4 bowls. "Purity" checks each bowl. If Bowl 1 is 90% apples, it's very pure. If Bowl 1 is a mix of apples, oranges, and bananas, it has low purity. We want each computer-generated group to contain mostly one language.

Algorithm	Silhouette Score	Cluster Purity
K-Means	0.2281	0.4361
Hierarchical Clustering	0.2119	0.4431

Table 8: Clustering Evaluation Metrics

5.2.2 Cluster Composition

We visualize how samples were distributed across the 4 clusters for both algorithms to understand the grouping logic.



(a) K-Means Cluster Composition

(b) Hierarchical Cluster Composition

5.2.3 Result Discussion

The analysis of cluster composition reveals an interesting underlying structure in the data. rather than a random distribution or a perfect language split, we observe distinct patterns:

- **The Stronger Signal:** The algorithms successfully found consistency, just not based on Language. Mathematical clustering is greedy it grabs onto the most obvious differences

first. In audio processing, the difference between a high-pitched voice (typically Female) and a low-pitched voice (typically Male) is numerically massive compared to the subtle difference between a German 'a' and a Spanish 'a'.

- **Natural Grouping:** The data naturally gravitated towards these acoustic properties (Pitch, Tone, Recording Environment). For example, Cluster 0 and 1 captured the majority of Romance language samples (Italian/Spanish), while Clusters 2 and 3 captured Germanic/Asian samples. This suggests the model did find linguistic similarities (language families) or shared prosodic features, but they were heavily modulated by speaker characteristics.

The unsupervised algorithms worked correctly they grouped similar-sounding files. It simply turns out that similar-sounding in raw audio features is defined more by the speaker's physical characteristics than the language they are speaking. This confirms the necessity of Supervised Learning (labels) to force the model to look past the loud features of gender and focus on the quiet features of phonemes.